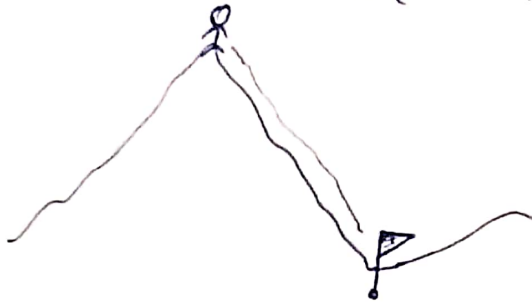


* Optimization in AI/ML.

Core idea:-

We want the model to learn parameters that minimize the loss (error)



Minimize Loss $\rightarrow$ Find best parameter

$$W_{\text{new}} = W_{\text{old}} - \eta \nabla L(W)$$

W = weight

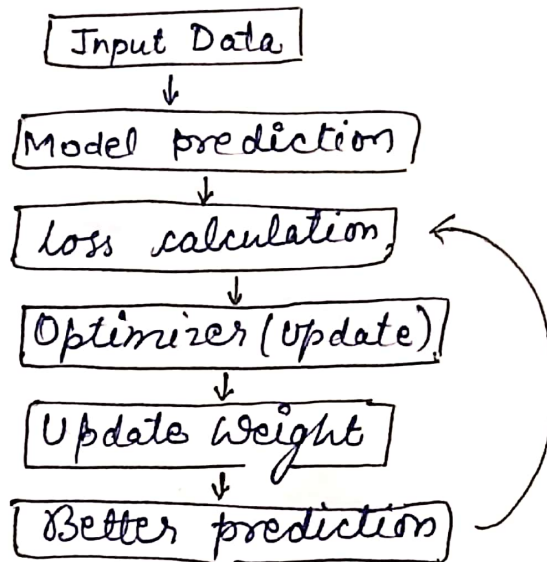
b = bias

L = loss function

η = learning rate

∇L = Gradient

Optimization pipeline



@codingdidi
Gradient descent

* Move in opposite direction of Gradient.

* Gradient points in direction of maximum increase.

Types

$\hookrightarrow$ Batch GD

$\hookrightarrow$ Stochastic GD

$\hookrightarrow$ Mini Batch GD

Common optimizer.

SGD $\rightarrow$ Basic, Noisy

Momentum $\rightarrow$ Faster less oscillation

AdaGrad $\rightarrow$ Adaptive LR per parameter

RMSprop $\rightarrow$ Fixes AdaGrad decay.

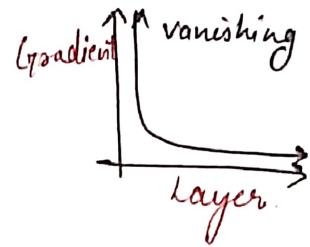
Adam $\rightarrow$ Momentum + RMSProp.

AdamW $\rightarrow$ Adam + Weight Decay
(Most Used in Deep learning)

Common issues & solution

↳ Vanishing Gradient

↳ Use ReLU, Batchnorm, Residual



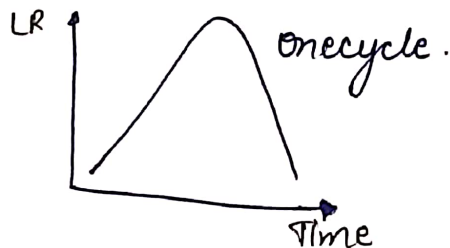
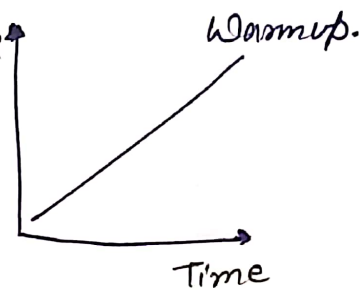
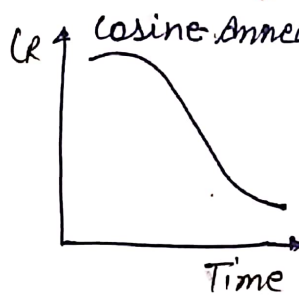
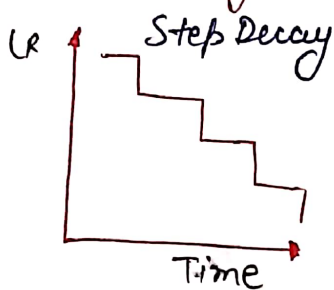
↳ Exploding Gradient

→ Gradient clipping

↳ Training too slow

→ Better optimiser, LR scheduler.

Learning Rate Scheduling ^{@codingdoshi}



① Too small → Training extremely slow.

② Too large → Model will overshoot the minimum

③ Ideal → Balanced Learning rate

Long term Memory vs.



Store information for a long duration from hours to a lifetime.

Example :- ~~a phone no.~~

→ ~~a list of groceries.~~
~~road~~

Short term Memory



Store information for a short duration from seconds to about a minutes.

